

(a) physical coordinates

mate by closest-point search

body A

$$\mathbf{t} = t_N \mathbf{n} + \mathbf{t}_T$$

$$\mathbf{x} = \varphi_A(\theta_A)$$

closest-point
search (iterative)

$\mathbf{n}$

g_N

$$\varphi_B(\psi_B)$$

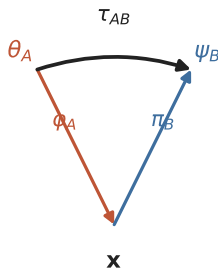
$\partial\Omega_A$

$\partial\Omega_B$

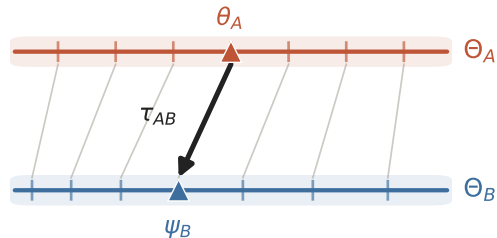
body B

$$\tau_{AB} = \pi_B \circ \varphi_A$$

evaluate φ_A , invert φ_B : no search



(b) reference coordinates



τ_{AB} transports Θ_A onto Θ_B (non-uniform reparametrisation)

(c) pulled-back gap $\bar{g}(\theta_A) = g_N^B(\varphi_A(\theta_A))$

